
THE PAPER'S STANDING WITHOUT THE CERTIFICATE

*A One-Page Companion to A Formal Proof of the Real Part of the Riemann Hypothesis, v2.0.2 ·
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What the kernel proves, what it imports, and the one sentence that would fall without the certified height

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ABSTRACT. The paper grades every result: **K**, a kernel theorem with no user-declared axiom; **K given A**, a kernel theorem whose hypotheses are cited theorems; **A**, a cited theorem itself. The certified height 3×10^{12} enters at grade A and populates one pre-proved schema. Were it absent from the literature, exactly one sentence of the paper would fall and every numbered theorem would stand. This handout states that in one page.

KEYWORDS Riemann Hypothesis; Real part; Unicorn part; certificate schema; grade K; grade A; Platt and Trudgian

The principle. The paper separates what the kernel proves from what it imports, and grades every result: **K**, a kernel theorem in core Lean 4 with no user-declared axiom; **K given A**, a kernel theorem whose hypotheses are cited theorems of the literature; **A**, a cited theorem itself. The certified height 3×10^{12} of Platt and Trudgian (2021) enters at grade A and populates one abstract, pre-proved schema. It is a parameter of that schema and a premise of no theorem.

The test. Suppose the result of Platt and Trudgian were absent from the literature. Exactly one sentence of the paper would fall: *the Real part of ζ is proved to height 3×10^{12}* ; and with it the Real part of ζ would stand proved at no height, the paper's own count to $T = 100$ being testimony, not a certificate. Every numbered theorem of Parts I, II, and III would stand unchanged, because none of them mentions a zero of ζ .

Four pillars that carry no zero of ζ . *The division law* (III.11): at any height T the hypothesis is exactly the Real part and the Unicorn part; *the fusion law* (III.1–III.6): fusing a proved part with an open part lowers the proved part to open, at $T = 3 \times 10^{12}$, at $T = 100$, or at $T = 0$ alike. *The schema and the arrow* (III.34, III.37–III.44): a complete on-line list proves the bounded universal, and the whole road holds with the Root Axiom replaced by the identity map, so no physical or empirical premise enters the deduction. *The seat* (Theorems A–C, C'): the scalar line is the fixed locus of quaternion conjugation, the identity cone closes at zero mass, and the chart of the critical line embeds equivariantly, at every resolution, without a zero height anywhere.

The equivalences and the asymmetry (Theorems 44–50, 68; E, F, F'; 70–77): Li positivity, $\Lambda = 0$, the transport of the modes, and Postulate M are one bit, given their imports; existence decides the line property on no frame; the hypothesis posits no object and only its denial posits a finite witness.

What the certificate does. It closes the Real part of ζ at one height, through the schema, as a cited theorem applied through a kernel theorem: grade A joined to grade K. The kernel checks the schema and the split, never the certificate's numerics, and the paper says so in the Standard of Proof, the Real Part section, and the title paragraph. Remove the certificate and you remove a height, not a theorem.

Every claim above is stated in the paper's Standard of Proof, where the same table appears; the audit log records the twelve external reports that tested it.

The paper's standing without the certificate. Each component, its standing, and whether it depends on the certified height.

Component	Standing	Depends on 3×10^{12} ?
The seat, its bound, the aperture, and the assignments of the three measures (Part I, Theorems A to I)	K, kernel theorems, no user-declared axiom; the physical reading of the price is a reading	No
The address and the cone, halted uniduction (Part II, Theorems 44 to 50 and 68)	K given A: Li's criterion, the $\Lambda = 0$ equivalence, the definitional identification, the seed	No
The logical form and the asymmetry of the denial (Theorems 70 to 77)	K given Σ_1^0 -completeness, and soundness in the converse only	No
The division at any height, the fusion law, the unicorn identity (III.1 to III.11)	K	No
The certificate schema, the placement, the bridge, the round trip, the arrow (III.12 to III.44)	K	No
The Real part of ζ to height 3×10^{12}	A joined to K, the cited certificate through the schema	Yes, and only this